Trevor Harris, PhD

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Statistics and machine learning researcher developing probabilistic models and uncertainty quantification methods for spatiotemporal data, with applications in climate, weather, and infectious disease.

Selected Research Projects

Conformal prediction and generative calibration

- Showed that differentiable conformal scores define analytic sampling flows. Mixing confidence levels yields calibrated predictive distributions and rare-event samples without training a separate generative model (NeurIPS 2026, accepted).
- Calibrated predictions for precipitation downscaling and climate debiasing with conformal predictive distributions. Reduced error in local spatial variability (MMD) by 75–78% versus the best of six neural baselines.
- Designed locally adaptive conformal inference for neural operators. Obtained 20% narrower bands than the functional conformal baseline with 92% functional coverage (90% target) on global ERA5 temperature forecasting with Fourier Neural Operators.
- Developed minimum distortion calibration via constrained optimal transport. Reduced raw deep ensemble calibration error by 91% and feature space energy score by 27% for shallow water forecasts.

Climate model integration, evaluation, and calibration

- Developed neural network Gaussian process regression to assimilate climate model output with observations. Reduced global temperature MSE by 58% versus ensemble averaging on held-out ERA5 data from 2015–2021.
- Co-developed spherical convolutional Wasserstein distance (SCWD) to detect regional distributional biases in global climate models. Evaluated 79 climate models on 2 TB of raw climate data and released SCWD in PyTorch (NeurIPS 2024 spotlight).
- Introduced conformal ensembles to calibrate climate projections against observations. Reduced average sliced Wasserstein error by 29–38% versus uncalibrated climate ensembles in tests on withheld models through 2099.

Graph neural networks for West Nile virus forecasting

- Co-designed GraphSAGE networks to forecast West Nile virus one week ahead. Achieved ROC AUC 0.844 versus 0.775 for XGBoost, the strongest tested baseline (GeoHealth, 2024).
- Co-developed GraphMAGE with skip connections from inputs to hidden layers to limit oversmoothing in deep GNNs. Improved mean F1 by 32% over GraphSAGE across eight forecast horizons (arXiv, 2025).
- Integrated 133,867 surveillance records, daily 4km weather, and 30m land cover data. Achieved the highest F1 among tested models on unseen urban and rural trap networks.

Technical Skills

Languages and frameworks. Python (PyTorch, JAX), R, SQL.
Methods. Conformal inference, graph neural networks, neural operators, generative modeling, optimal transport, Gaussian processes, spatiotemporal statistics.
Scientific data. ERA5, WeatherBench2, CMIP6, paleoclimate reconstructions, mosquito surveillance.

Education and Selected Honors

PhD, Statistics, University of Illinois Urbana-Champaign (advisor: Bo Li) 2021
MS, Statistics, University of Illinois Urbana-Champaign 2018
BS, Mathematics, University of Florida, *magna cum laude* 2014

Honors. JABES Best Paper (2024); Environmetrics top 10 most cited papers, 2022–2023 (2024); Norton Prize for outstanding Statistics thesis, UIUC (2021); UIUC nominee, Schmidt Science Fellows (2020).

Research and Professional Experience

University of Connecticut 2024–Present
Assistant Professor, Statistics (AI/ML Cluster)

- Develop conformal inference, generative neural operators, and distribution control for weather and climate. Train weather models at 1.5° resolution and U-Nets to estimate tail dependence coefficients at high quantiles on NSF ACCESS GH200 GPUs.
- Lead a $234K Sandia-funded research program on conformal uncertainty quantification (PI, 2024–2027).
- Advise PhD research on manifold-constrained conformal prediction and latent neural ODE climate emulation. Teach Deep Learning and Statistical Programming (Python).

Texas A&M University 2021–2024
Assistant Professor, Statistics

- Developed climate validation and projection methods, and graph neural networks for disease forecasting.
- Led a $300K Sandia-funded project assessing climate intervention outcomes and a $10K project modeling West Nile virus under extreme climate (PI).
- Supervised undergraduate and master's research. Served on PhD dissertation committees. Taught Machine Learning, Bayesian Statistics, and Computational Statistics.

Sandia National Laboratories Summers 2018, 2019–2020
Graduate Intern, Mission Algorithms R&S

- Researched functional time series changepoints and trajectory anomalies. Released R packages fmci and elasticdepth (Environmetrics and Technometrics).

GEICO 2014–2016
Product Modeling Analyst, Underwriting Research

- Built pricing and risk segmentation models using SQL and R on 100+ GB of policyholder data. Rebuilt a legacy modeling pipeline as an R package, automating a manual reporting workflow across teams.

Selected Publications and Manuscripts

1. **Harris, T.** (2026). *Flow-Based Conformal Predictive Distributions*. NeurIPS, accepted.
2. Garrett, R., **Harris, T.**, Wang, Z., & Li, B. (2024). *Validating Climate Models with Spherical Convolutional Wasserstein Distance*. NeurIPS (**spotlight**).
3. Tonks, A., **Harris, T.**, Li, B., Brown, W., & Smith, R. (2024). *Forecasting West Nile Virus with Graph Neural Networks: Harnessing Spatial Dependence in Irregularly Sampled Geospatial Data*. GeoHealth.
4. **Harris, T.**, & Sriver, R. (2024). *Quantifying Uncertainty in Climate Projections with Conformal Ensembles*. Annals of Applied Statistics, accepted.
5. **Harris, T.**, Li, B., & Sriver, R. (2023). *Multimodel Ensemble Analysis with Neural Network Gaussian Processes*. Annals of Applied Statistics.
6. **Harris, T.**, & Liu, Y. (2025). *Locally Adaptive Conformal Inference for Operator Models*. TMLR, under review.
7. Nill, C., **Harris, T.**, & Adams, J. (2026). *Manifold Constrained Conformal Prediction for Spatial Distributions*. ICLR, under review.

Additional publications: Google Scholar and research website.

Selected Professional Activities

Recent invited talks. CMU (2025, 2026); Penn State, University of Maryland, Verisk Analytics, SIAM UQ, JSM (2026); IMSI (2025); keynote at ICDS (2023).
Selected service. Associate Editor, JABES (2026–Present) and the AI+Climate special issue of ASCMO (2024–Present). Reviewer for NeurIPS, ICLR, JMLR, and JASA.